Exp 2-4 Part 2 Report P7

Hanwen Zhang, Qinyun Hu, Yuexiang Hou, Siyuan Yin, Feiyang Peng, Yiling Chen

1. Present Results

For the four possible influencing factors already listed in the Plan

1) The type of polymer
2) The amount of water
3) The setting temperature
4) The glass powder used in the cements

The factorial designed experiments were used to test the effects of all four factors at the same time. Each factor is set at two levels using symbols + and - to distinguish, and the details can be seen in Table 1.

Table 1 Factors and levels used in factorial designed experiments.

Factors		Levels	
		+	-
A	Polymer	P1 (old polymer)	P2 (new polymer)
B	Water/ Solvents	1:7	1:9
C	Temperature	35°C	20°C
D	Glass powder	G1 (from old glass powder supplier)	G2 (from new glass powder supplier)

For four variables, 16 sets of runs should be designed, for each run making 5 turns. The next step is to calculate which factors, or combination of factors, have actually caused the new material to change its properties.

Table 2 Table of possible factors and interactions.

Factors	Interactions
A, B, C, D	AB, AC, AD, BC, BD, CD

The level of the interactions follows the rule,

Table 3 Rule of level of the interactions.

Factor A	Factor B	AB
+	+	+
+	-	-
-	+	-
-	-	+

The following table shows the notation of the parameters in the calculation.

Table 4 Parameters in the processing and related formula.

Notations	Descriptions	Formula
N	Number of values for each sum	/
Y	Outcomes	/
Y'	The mean outcome	$\frac{Y_1 + Y_2 + \ldots + Y_n}{n}$
R	The range of Y	$Y_{max} - Y_{min}$
R'	The mean range of Y	$\frac{R_1 + R_2 + \ldots + R_n}{n}$
SUM+	Sum of Y'/R of '+' factor	/
SUM-	Sum of Y'/R of '-' factor	/
Diff	Difference of SUM+ and SUM-	|(SUM+)-(SUM-)|
Effect		$\frac{\lvert Diff \rvert}{N}$
/	Random variation	$\frac{(R' \times 2)}{N}$

The processing of the calculation is carried out as follows:

1. Calculating the Effect (Y') and R for each turn, total Y and R', and recording them at the right of the table.
2. Calculating the sum+ and sum- of Effect (Y') and R for each factor and recording it at the bottom.
3. Calculating Differ and Effect of Effect (Y') and R for each factor and recording it at the bottom.
4. Comparing the every effect value with random variation, if the effect is much bigger, then the factor is really related to the change in the properties of new materials.

The result of the calculation are shown in ' Experiment Data - GIC '.

2. Analyse Results

In this factorial-designed experiment, Effect(Effect (Y')) is used to measure product test performance and Effect(R) is used to measure the reliability of usage. By analysing these two parameters together to measure the impact of factors, either individually or in pairs, on a product and to analyze the relationship between the impacts of factors, which are clearly shown in Table 5.

Table 5 Effect(Y') and Effect(R) of different factors (Random variation=9.66).

Factors	Polyemr	Water	Temperature	Glass						
	A	B	C	D	AB	AC	AD	BC	BD	CD
Effect (Y')	45.04	16.53	4.97	2.05	25.20	21.17	2.30	12.58	1.14	0.06
Effect (R)	47.16	11.66	1.79	10.39	3.81	23.24	5.49	21.34	4.79	2.46

The parts marked in red are much greater than Random variation, which is $\mathrm{Y'}$ or $\mathrm{R} \geq 1.5 * \mathrm{Random\ variation}$.

2.1 The analysis of Effect(Y')

(√ : influential factor; × : non-influential factor)

Table 6 The influential/non-influential factors and interaction between each two single factors.

Factor	A	B	C	D
A	√	√	√	×
B	√	√	×	×
C	√	×	×	×
D	×	×	×	×

2.1.1 The analysis of Effect(Y') on A/B/AB

Both the type of polymer (A) and the amount of water added (B) significantly affect product behavior, with their Y' being much larger than Random variation. A has a greater influence than B, highlighting its dominant role. The interaction between polymer type and water amount (AB) also emerges as an influential factor, suggesting a notable interaction between A and B. This interaction may occur because water causes the carboxyl groups (-COOH) in polyacrylic acid to partially dissociate into hydrogen ions (H+) and carboxylate ions (-COO-).

2.1.2 The analysis of Effect(Y') on A/C/AC

When considering the Effect (Y') for temperature (C), it becomes clear that temperature alone is not a significant factor, as its Effect (Y') is less than Random variation. However, the interaction between polymer type and temperature (AC) shows an Effect (Y') greater than Random variation, proving that AC is influential due to this interaction. This interaction is reflected that different types of polyacrylic acid vary in temperature sensitivity, with high temperatures potentially degrading their properties.

2.2 The analysis of Effect(R)

(√ : reliable; × : non-reliable)

Table 7 The reliable/non-reliable factors and interaction between each two single factors.

Factor	A	B	C	D
A	×	√	×	√
B	√	√	×	√
C	×	×	√	√
D	√	√	√	√

2.2.1 The analysis of Effect(R) on AB

AB: The Effect(R) of polymer type (A) greatly exceeds Random variation, making it unreliable, while the Effect(R) of water amount(B) is similar to Random variation, making it moderately reliable. However, their combination (AB) has a much smaller Effect(R) than either A, B or even Random variation, demonstrating high reliability and a significant interaction between polymer type and water amount.

2.2.2 The analysis of Effect(R) on AD

D/AD: For the type of glass added (D), the Effect(R) is close to Random variation, indicating moderate reliability. The interaction between polymer type and glass type (AD) shows a smaller Effect(R) than Random variation, ensuring high reliability and indicating an interaction, which is not the same as the previous analysis.

2.2.3 The analysis of Effect(R) on BD

B/BD: The analysis of water amount and glass type (BD) results in an Effect(R) smaller than Random variation, demonstrating high reliability and interaction. This interaction may lie in when different types of glass powders require different amounts of water to form an optimal gel and cure structure.

2.2.4 The analysis of Effect(R) on C/CD

C/CD: The interaction between temperature and glass type (CD) is not so notable, which indicates high reliability.

In conclusion, polymer and water both interact together with temperature making the problem much more complex.

3. Discuss the Conversation

Comparing people's ideas with the above calculations.

3.1.1 Thinking about replacing the polymer

The thinking about replacing the polymer **(Simon (R&D): Wasn't that about the time we changed the polymer?)** point out that Simon aware that polymers undergo changes due to their beneficial properties but unaware of the underlying cause of the problem.

3.1.2 Thinking about replacing the glass ions

The thinking about replacement of the glass ions **(Laura: We changed the supplier of our glass powder1.)** observes that the creation of materials is contingent upon the mixing of polymers and the occurrence of breakthroughs in water. Consequently, alterations in the polymers will result in alterations in the molecular patterns that make it challenging to mix the materials together, thereby rendering the mixture unreliable.

3.1.3 Thinking about the different amount of water

The thinking of adding **(Darren (Rheology technologist): "Whenever I find the mix is difficult to work, I just add a bit more water.)** merely points out what to do without knowing why.

3.1.4 Thinking about the change the temperature

The thinking about change the temperature **(Simon: But you do all your testing with samples set at room temperature. The dentist sets them in the patient's mouth at 35°C.)** notes that the filler was synthesised by combining the polymer and glass powder in an aqueous environment, indicating that temperature was not a significant factor in initiating the reaction.

3.1.5 Thinking about product batches

In the discussion on product batches (Laura: "glass supposed to have exactly the same specifications as the old one.") notes the differences in glass product batches, but doesn't know the possible effects. The company did not check whether the glass powder altered the processing conditions, nor did it understand the impact on other aspects of the process.

4. Compare with Design

In the experimental plan, the group set the following factors:

1) The changed raw polymer；
2) The concentration of the filling；
3) The testing temperature different from the temperature in the mouth;
4) The changed supplier of glass powder.

And in factorial experiment the following factors were experimented and results were obtained:

Table 8 Factors experimented with in factorial experiment.

Factors		**Levels**	
		+	-
Polymer	A	P1	P2
Water	B	HIGH	LOW
Temperature	C	35	20
Glass	D	G1	G2

From the above, the factors to be experimented on in the factorial experiment were also set up as factors to be tested in the experiment plan. At the same time, a series of experiments, such as the Compression test, were also set up in the group's experimental plan to test the effect of factor changes on performance.

However, the data collection form of the experiments planned by the group was slightly different from that of the factorial experiment. The Compression test in the experimental plan only collected data on whether the sample was damaged or not, but not the Compressive strength, which may have led to the experiments in the experimental plan not obtaining similar results to the factorial experiment when it was finally carried out, resulting in a different answer to the factorial experiment.

However, the group experimental plan has the same underlying logic as the factorial experiment, so the group's experimental plan can also get the effect of each factor on the final performance.

Since the factors set up in the group experiment plan were the same as in the factorial experiment, the group was also able to find all the factors that changed the process based on the experimental plan

In the experimental plan, the group set up the following controlled experiment to investigate the effects of each factor:

Table 9 Experiments set up in the experimental plan.

Test	**NO.**	**Condition**			
		Polymers	**Concentration**	**Glass powders**	**Set temperature**
1	1-1	old	2g/ml	type 1	35°C
	1-2	new	2g/ml	type 1	35°C
2	2-1	old	2g/ml	type 1	35°C
	2-2	old	1g/ml	type 1	35°C
3	3-1	old	2g/ml	type 1	35°C
	3-2	old	2g/ml	type 2	35°C
4	4-1	old	2g/ml	type 1	35°C
	4-2	old	2g/ml	type 1	25°C

Compared to the factorial experiment, it shows that the group's experimental plan only experimented on the change of individual factors and not on the change of combinations between the factors, so it is unfortunate that the group was not able to find out the interactions between the factors through the experimental plan.

Compared to the factorial experiment, the group's experimental plan tested fewer samples but conducted more multiple experiments. Because the different kinds of experiments can be conducted simultaneously, the group's experimental plan will consume less time than the factorial experiment. However, due to the increase in the total number of experimental groups conducted, the group experimental plan will consume more human and material resources.

5. Future Plan

In a follow-up, the team will first contact the customer to inform about possible problems with the product. The second step is to carefully investigate the relationship between polymers and water versus temperature: use the new polymer to test mixtures with different amounts of water to understand water sensitivity. Finally, temperature control during testing will be improved.

According to the data analysis, when a single factor is considered, the polymer and water have an impact on the filling material, and the product reliability is low. When combining the two factors, it is found that any combination of polymer, water, and temperature will affect the performance of the filling material, but the product performance determined by the two factors of polymer and water has been stable. Therefore, to ensure high quality, companies should deeply explore the effects of polymers and water. At the same time, to ensure the stability of the product at the same time comprehensive consideration of two factors.

Based on the above results, to obtain high-quality and stable products, the core of the company's next research should be to select the right type of polymer and explore the performance of the product under different water amounts. The selection of suitable polymers can be considered from the aspects of molecular characteristics, reaction process, reaction conditions, and so on. Adding water needs to be explored experimentally.

After the suitable polymer is selected, the experiment is carried out at an oral temperature, and the samples are then tested for pressure resistance by setting up water with different concentration gradients. In the end, the company can determine the most appropriate polymer choice at the mouth temperature and the amount of water required, resulting in a high-quality, stable product.

6. Reference

1. Exp2-4 Experimental Plan P7.

2. Experimental data - GIC.